

Pythagoras, mensuration and 3D - Set 1

TMUA practice questions by topic. Specification M5.7, M5.11-M5.13, M5.15-M5.17. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its longest diagonal.
- A $\sqrt{153}$ cm
 - B 13 cm
 - C 15 cm
 - D 5 cm
 - E $\sqrt{119}$ cm
 - F 12.5 cm
- 2 A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.
- A 14 cm
 - B 6 cm
 - C 7 cm
 - D 11 cm
 - E 3.5 cm
 - F 5.5 cm
- 3 A cone has base radius 6 cm and slant height 10 cm. Find its volume. (Volume of a cone $= \frac{1}{3}\pi r^2 h$.)
- A $60\pi \text{ cm}^3$
 - B $72\pi \text{ cm}^3$
 - C $288\pi \text{ cm}^3$
 - D $48\pi \text{ cm}^3$
 - E $96\pi \text{ cm}^3$
 - F $120\pi \text{ cm}^3$

- 4 A sphere and a cube have the same volume. The cube has side length a . Find the radius of the sphere. (Volume of a sphere $= \frac{4}{3}\pi r^3$.)

- A $\frac{a}{\sqrt[3]{\pi 4\pi}}$
 B $\frac{a}{\sqrt[3]{\frac{4\pi}{3}}}$
 C $\frac{2a}{4\pi}$
 D $\frac{4\pi}{\sqrt[3]{\frac{3}{4\pi}}}$
 E $\frac{a}{\sqrt[3]{\frac{4\pi}{3}}}$
 F $\frac{a}{\sqrt[3]{4\pi}}$

- 5 A square of side 10 has a quarter-circle of radius 10 drawn inside it, centred at one corner. Find the area of the square not covered by the quarter-circle.

- A $100 - 50\pi$
 B $100 - 25\pi$
 C $100 - 100\pi$
 D $100\pi - 100$
 E $50 - 25\pi$
 F 25π

- 6 A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?

- A 7 m
 B 5 m
 C 4 m
 D 5.5 m
 E 6 m
 F $\sqrt{48.5}$ m

7 In a right-angled triangle the hypotenuse is 10 and one angle is 30° . Find the area of the triangle.

- A $\frac{25}{2}$
- B 25
- C $\frac{25\sqrt{3}}{2}$
- D $\frac{25\sqrt{3}}{4}$
- E $\frac{50\sqrt{3}}{25\sqrt{3}}$
- F $\frac{25\sqrt{3}}{4}$

8 The volume of a cube is numerically equal to its surface area. Find the length of its side.

- A $\sqrt{6}$
- B 3
- C 2
- D 12
- E 1
- F 6

9 Which of the following statements is/are true?

- I Doubling the radius of a circle doubles its circumference.
- II Doubling the radius of a sphere multiplies its volume by 8.
- III Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

- 10** A running track consists of two straights of length 100 m and two semicircular ends of radius r . The total length of the track is 400 m. Find r .
- A** 100π
B $\frac{100}{\pi}$
C $\frac{200}{\pi}$
D $\frac{400}{\pi}$
E $\frac{300}{\pi}$
F $\frac{500}{\pi}$
- 11** A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?
- A** 6 cm
B 12.5 cm
C 8 cm
D 7.2 cm
E 5 cm
F 6.4 cm
- 12** A solid consists of a hemisphere of radius 3 on top of a cylinder of radius 3 and height 4. Find its volume. (Sphere volume $\frac{4}{3}\pi r^3$.)
- A** 90π
B 45π
C 63π
D 36π
E 72π
F 54π